



Follow-up Observations and an Improved Orbital Period for Exoplanet Candidate TOI-3505.01

Mason Cao¹, Annalyse Dickinson², Owen Alfaro², Rianne Eccleston², Kasey Davidson², Peter Plavchan²

¹Lambert High School, Suwanee, Georgia; ²Mason Space Center, Department of Physics and Astronomy, College of Science, George Mason University

Schar Scholars three month internship:
NASA Data Science Analysis and Scholarly Research Paper Writing

Abstract

TOI-3505.01 is a planet candidate found by NASA's TESS mission. It circles a 6220 K star that is 1.34 times the radius of the Sun (TIC 390988385, $V = 11.2$) every 2.9151 days. On July 22, 2022 we watched the star with the 0.8 m telescope at George Mason University during the hours the transit was supposed to happen. We took 283 images in R band and kept 263 good measurements, with a scatter of 2.50 ppt. The star never dimmed (-0.66 ± 0.39 ppt). To see whether that meant anything, we added fake transits to our data. A fake 2.91 ppt transit came back at 5.7 σ , so we would have caught a real one.

We saw nothing because the prediction was off. With the period we have now, the closest transit was 17.5 hours before we started. So we went to the TESS archive and timed 27 transits in Sectors 14, 41, 54, and 81, a 5.05 year stretch. Fitting a line through those times gives **$P = 2.9151516 \pm 0.0000049$ days**. That is 2.4 times tighter than the TESS catalog period and 1.4 times tighter than the best published one, and it agrees with both. It also cuts the predicted transit time for 2026 from 9 minutes to 3.

Introduction

TESS watches one strip of sky at a time and looks for stars that dim on a schedule, which is what a planet passing in front of one does (Ricker et al. 2015). It has found thousands of candidates and most are still only that. Ground telescopes do the next step: they check the dimming belongs to the star TESS blamed, at the depth and time predicted, and not to a fainter star nearby whose light lands in the same pixels.

TOI-3505.01 has been in that queue since 2021 and nobody has published on it. Six observatories have taken light curves and the TESS project still calls it a candidate. It looks like a Jupiter-sized object grazing a star a little bigger and hotter than the Sun. Here we present our own night of R band photometry and a new timing analysis of every TESS sector that has observed it.

Materials and Methods

Taking and measuring the images

- Mason 0.8 m telescope, July 22, 2022 UT. R filter, 50 second exposures, 0.362" per pixel. We got 283 images and 281 of them lined up into one 5.07 hour run.
- We subtracted darks, divided by flats, and shifted every image so the stars matched, using 25 reference stars, then redid each step in Python to check it. Astrometry.net matched our field to sky coordinates and landed 0.47 px from our measured position.
- We measured brightness in AstrolmageJ with a **25 px (9.1") aperture**, a **70–139 px sky ring**, and comparison stars C2–C11. We picked 25 px by trying every radius from 20 to 50 px and keeping the one with the least scatter. Our own Python photometry agrees with AstrolmageJ at $r = 0.990$.
- We cut 18 measurements for bad image quality and wrote down why for each, leaving **263**. No detrending passed the test we set up ahead of time, so the light curve is left as measured.

Working with the TESS archive

- We used QLP for Sectors 14 and 41, the only pipeline covering them, and the SPOC 2-minute curves for Sectors 54 and 81. Both pipelines measure the same photons, so only the better-sampled one speaks for each sector. We also pulled every Data Validation report.
- We ran a box least-squares search per sector, then fit each transit one at a time for 27 mid-transit times and fit a straight line through them. The shape we fit is a trapezoid, not a flat box: at $b = 0.92$ this transit spends most of its time in ingress and egress, and a box has edges too sharp to describe that.
- We dropped one Sector 81 event that straddles a data gap. It fills under half the fit window, and with the middle missing the baseline and the depth stop being separable.
- We used the TESS pixel data to build difference images, find where on the sky the dimming happens, and check how much light nearby stars add. We locked in our cuts before looking at the final numbers.

Checking the neighbours

- TFOP asks for clearance inside 2.5', so we measured all 280 catalog stars in that circle bright enough to matter, on our own images, across the whole night. A star clears when our depth limit for it falls below the eclipse it would need.

Checks for False Alarms

Same depth in all four sectors	$\chi^2 = 4.3 / 3$
Odd and even transits match	$\leq 1.7\sigma$
No second dip half an orbit later	$S/N \leq 1.7$
Dimming centered on our star	1.6–5.1"
Compared with one TESS pixel	inside 21"
Nearby stars ruled out, 2.5'	222 of 280
SPOC odds of a false alarm	1×10^{-143}
Star 0.517" away	cannot separate

Future Work

- Watch a transit using our new period. A 3 minute window makes that worth doing on a 0.8 m telescope.
- Fit a real limb-darkened model with MCMC and a density prior to pin down b and the planet's size.
- Pull the four-colour photometry and the six Keck velocities other groups already filed and test them against this geometry. A depth that changes with colour would mean a blended eclipsing binary.

Results

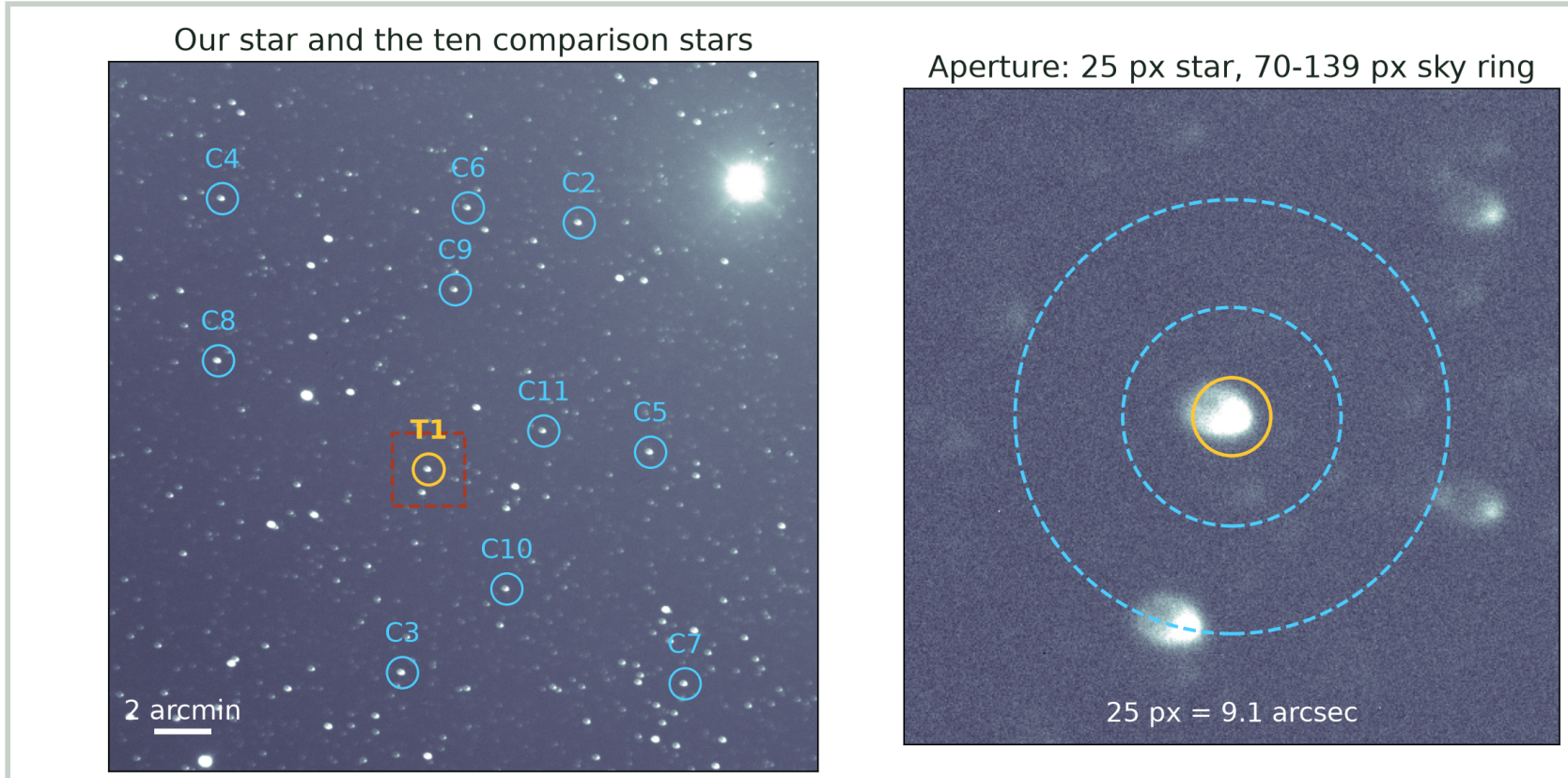


Fig. 1. The field around TOI-3505 after we matched it to sky coordinates. T1 is our star and C2–C11 are the comparison stars. The right panel zooms in on the 25 px aperture and the 70–139 px sky ring we used.

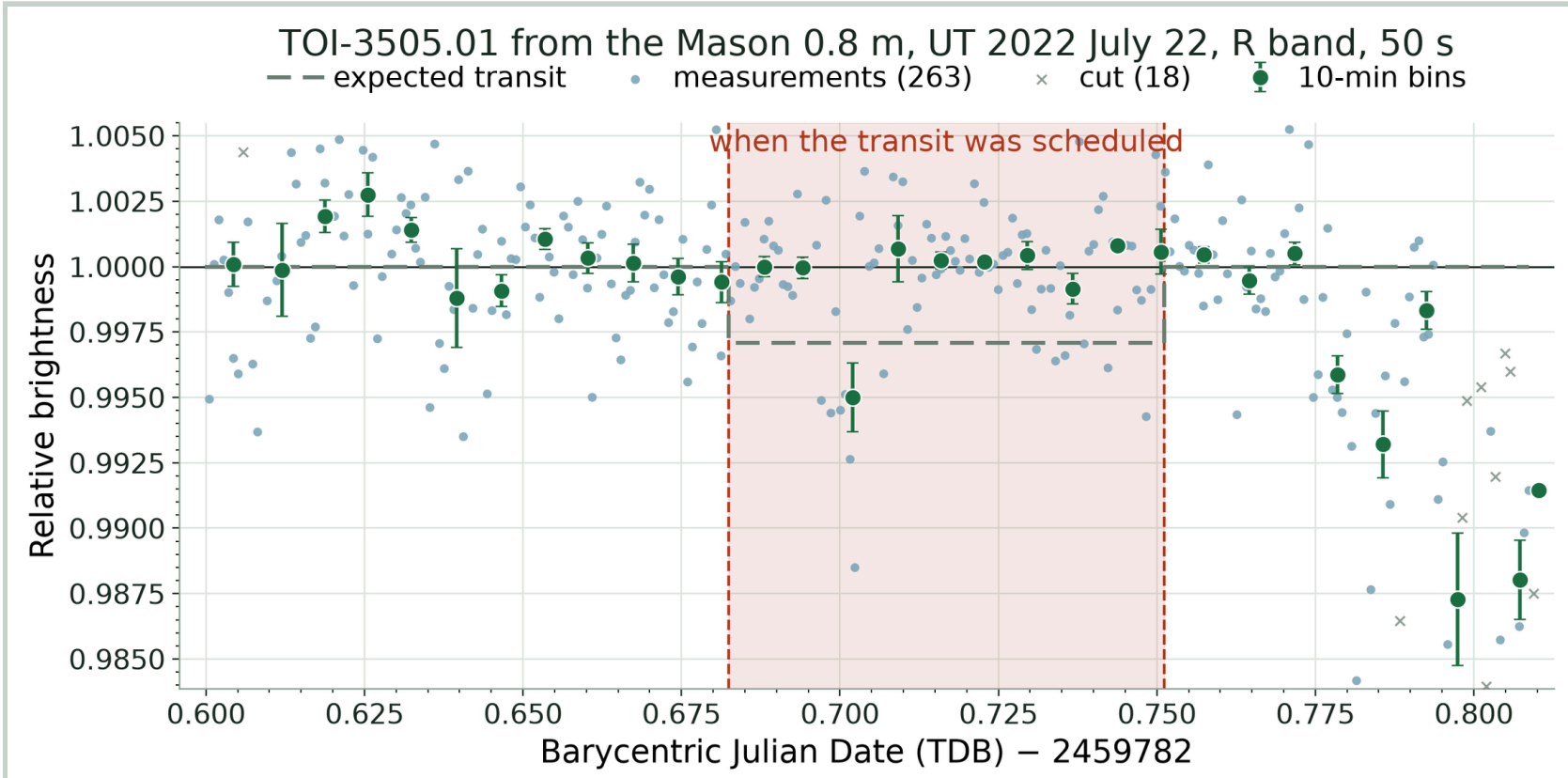


Fig. 2. Our final light curve: 263 measurements, 2.50 ppt scatter, no detrending. The dashed line shows the dip we would have seen if the transit had happened on schedule. Nothing is there. The points get noisier at the end of the night as the star dropped closer to the horizon.

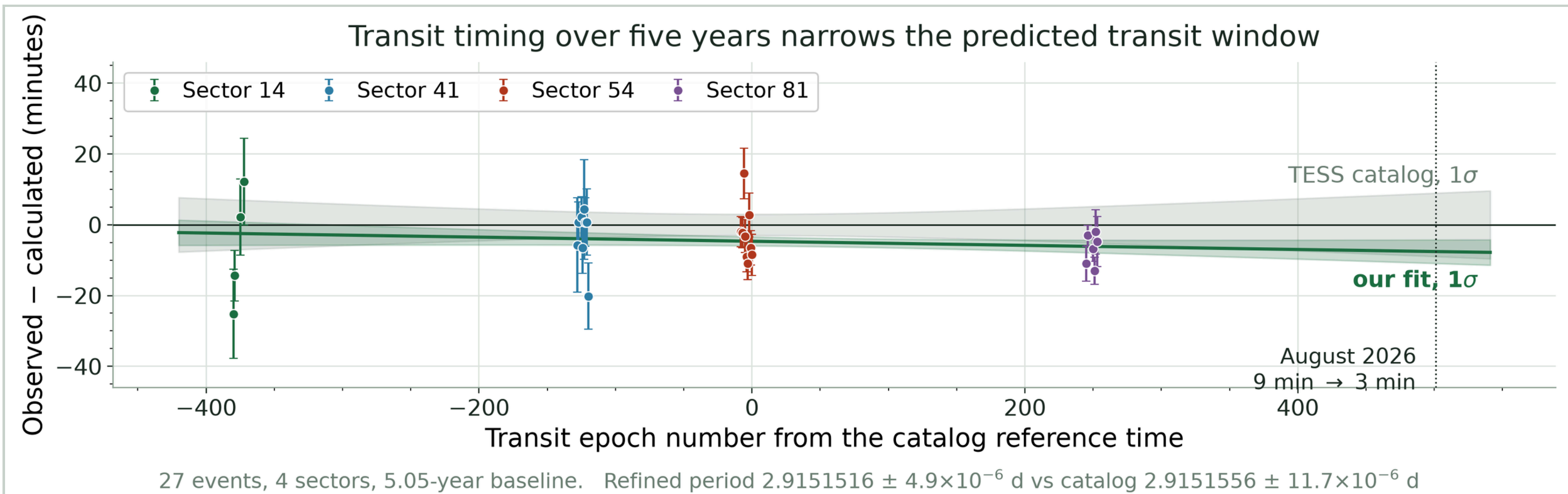


Fig. 3. Our main result. Each point is one transit we timed, compared with the time the TESS catalog predicts. Grey is the catalog's 1 σ range and green is ours. Our transits stretch across 5.05 years, which is the whole reason the period tightens, and the predicted transit time for 2026 goes from 9 minutes to 3. We widened the error bars by 1.14 first, so this is the careful version.

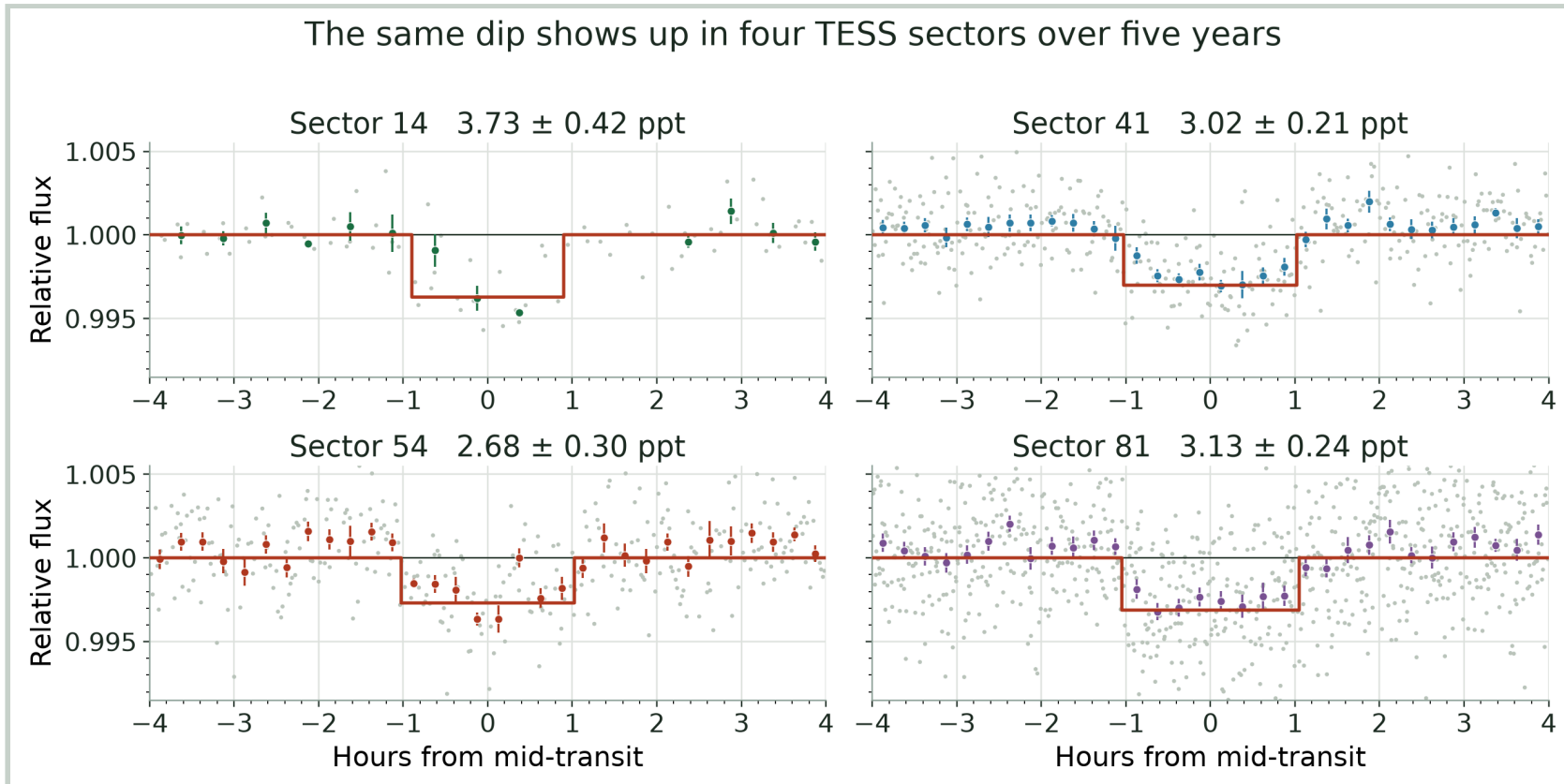


Fig. 4. TESS data folded on our new period. The dip shows up on its own in all four sectors across five years, and the four depths agree ($\chi^2 = 4.3$ for 3 degrees of freedom).

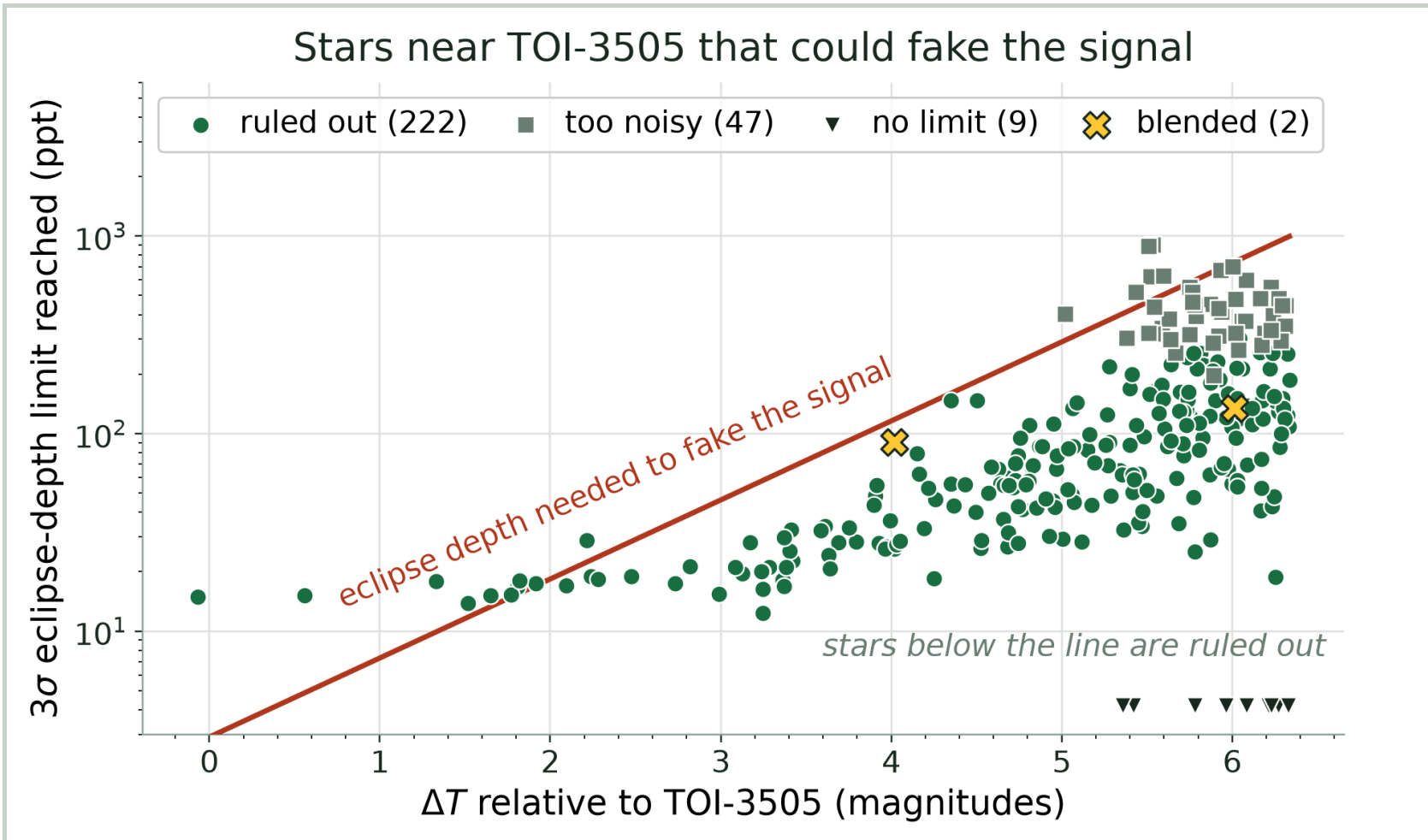


Fig. 5. All 280 catalog stars inside 2.5' that are bright enough to fake our signal, the radius TFOP asks for. The red line is the eclipse depth each one would need. A star is ruled out when its own light curve is good enough to say that eclipse did not happen. 222 are. The rest are too faint to measure well, or sit inside our aperture.

Our numbers next to the published ones

Quantity	Ours — S14, 41, 54, 81	TESS catalog — S14, 41, 54	SPOC — S54, 81
Orbital period (days)	2.9151516 ± 0.0000049	2.9151556 ± 0.0000117	2.9151456 ± 0.0000070
Years of transits behind it	5.05	3.0	2.0
Transit depth (ppt)	3.05 ± 0.13	2.91 ± 0.20	3.29 ± 0.12
Transit length (hours)	2.06 ± 0.05	2.00 ± 0.21	2.71 ± 0.10

Depth and length are our own four-sector fits. Sector counts are the transits behind each period: ours span 5.05 years, the catalog row three sectors, SPOC two.

Major Citations

- Ricker, G. R. et al. 2015, *JATIS*, **1**, 014003.
- Collins, K. A. et al. 2017, *AJ*, **153**, 77.
- Reefe, M. et al. 2022, *JATIS*, **8**, 027002.
- Huang, C. X. et al. 2020, *RNAAS*, **4**, 204.
- Jenkins, J. M. et al. 2016, *Proc. SPIE*, **9913**, 99133E.
- Twicken, J. D. et al. 2018, *PASP*, **130**, 064502.
- Mandel, K. & Agol, E. 2002, *ApJ*, **580**, L171.
- Stassun, K. G. et al. 2019, *AJ*, **158**, 138.

System Parameters

The star, TIC 390988385	
Temperature	6220 K
Radius, assumed mass	1.335 R _☉ , 1.25 M _☉
Distance, TESS mag, V	373 pc, 10.94, 11.24
Nearby star, extra light	0.517" ($\Delta I = 1.7$), 0.55
What that gives for the planet	
R _p /R _★	0.0618 ± 0.0016
Planet radius	9.0 R _⊕ = 0.80 R _J
How centered, tilt	b = 0.85, 82.9°
a/R _★ , orbit, temperature	6.93, 0.043 au, 1670 K

From the TIC v8 star radius and the SPOC depth ratio, with no correction for the nearby star. Reference time BJD_{TDB} 2459793.5311 ± 0.0009.

Discussion

Why we saw no transit

We picked this night using a 2022 prediction. With the period we have now, the nearest transit was 17.5 hours before we started, so there was nothing to catch. Our run also lands in a gap in TESS Sector 54, so we cannot line the two up. We added fake 2.91 ppt transits to our data and pulled them back out at 5.7 σ , so a real transit would have shown up. Our flat light curve is a real limit and not just noise.

What the TESS archive gave us

One night could not fix the prediction, so we used the archive instead. Timing 27 transits across four sectors gives **$P = 2.9151516 \pm 0.0000049$ d**. What buys the precision is the lever arm. Our transits span 5.05 years, while the catalog row rests on three sectors and the official SPOC fit on two, so we come out 2.4 times tighter than the catalog and 1.4 times tighter than SPOC while agreeing with both (0.3 σ and 0.7 σ). Scatter about the line is 8.5 min for $\chi^2/\text{dof} = 1.30$, so we widened the error bars by 1.14. Dropping Sector 14, where each image covers 30 minutes, moves the period by 0.6 σ .

Why the planet's size is uncertain

In the official SPOC fit the planet barely clips the edge of the star ($b = 0.92$). For that fit to work, the star would have to be 3.6 times less dense than a 1.34 R_☉, 1.25 M_☉ star can be. When a transit is this close to the edge, the depth, the length, b , and $a/R_{\star}$ all trade off, so lots of combinations fit the same data. If we keep the star's density and match our own 2.06 h transit length, we get $b = 0.85$ and $a/R_{\star} = 6.93$. Either way the planet only grazes the star, so 9.0 R_⊕ is much less certain than its error bar makes it look.

What we cannot claim

Another star sits 0.517" away and adds about 21 percent of the primary's light. Adaptive optics on the Shane 3 m sees it at $\Delta J = 1.53$ and $\Delta K_s = 1.59$, so it is redder than our target. Our telescope cannot split the pair and neither can TESS, so if the transit is on that star the depth and radius are bigger than we quote. Our SPOC comparisons use the same TESS data we did, so they check our method rather than add a detection. We are also not the first to work on this star. TFOP already holds six other ground light curves, imaging, and five spectra, none of it published. What nobody had done is fit an ephemeris across all four sectors.

Conclusions

- A better period.** Timing 27 transits over 5.05 years gives $P = 2.9151516 \pm 0.0000049$ d, tighter than both published values and agreeing with each.
- Better predictions.** The transit time for 2026 is now good to 3 minutes instead of 9, so the next person to observe needs much less time on either side of it.
- Our night still counts.** The 2.50 ppt light curve rules out a transit during the window we watched, and our fake-transit tests show we would have seen one.
- Still a good candidate.** Nothing we checked points to a false alarm. The signal looks like a roughly Jupiter-sized planet grazing the edge of its star.
- What holds us back.** The grazing path and the nearby star limit the size measurement now, not our photometry.

Acknowledgements

This research was made possible through the support of George Mason University's College of Science, which supports the Schar Scholars Program. We used data from NASA's TESS mission, which we got through the Mikulski Archive for Space Telescopes, and data from ESA's Gaia mission.